

AVIJIT PAKHIRA

Kolkata, West Bengal, India

+91 7044419685 avijitpakhira2807@gmail.com linkedin.com/in/avijit-pakhira-8541841ba github.com/avijit-28

Professional Summary

Aspiring Machine Learning Engineer and Data Scientist with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and NLP. Skilled in Python, TensorFlow, PyTorch, Scikit-learn, and OpenCV, with experience developing end-to-end AI solutions, conducting research, and building data-driven applications.

Education

JIS University

Bachelor of Technology in Computer Science (AI/ML) | CGPA: 9.36 / 10.0

Sep. 2022 – May 2026

Kolkata, West Bengal

Experience

Edunet Foundation (TechSaksham — Microsoft SAP CSR)

March 2025 – April 2025

Machine Learning Intern

Virtual

- Developed an SVM-based fitness tracking application using Python, Streamlit, and Scikit-learn, processing 15,000+ health records across 4 model comparisons to achieve 97.33% accuracy (Precision: 0.97, Recall: 0.96, F1-Score: 0.96)
- Engineered an end-to-end ML pipeline — data preprocessing, feature engineering, model training, and deployment — using Pandas, NumPy, and Scikit-learn, benchmarking 4 algorithms on 15,000 records to select the optimal SVM model
- Designed an interactive Streamlit dashboard delivering personalized fitness insights and real-time calorie predictions for users.
- Performed statistical analysis of model performance metrics (accuracy, precision, recall, MAE: 3.59, MSE: 26.37, RMSE: 5.14) to fine-tune hyperparameters and improve prediction reliability

Research & Publications

Ancient Vedic Mathematics with Karatsuba Multiplication for Efficient Face Recognition 2025-2026

Published a Springer-indexed research paper in AI and Computer Vision.

- Co-authored a Springer Nature research paper proposing a Vedic Cosine Similarity (VCS) framework by integrating FaceNet, MTCNN, and Karatsuba multiplication for efficient face recognition.
- Designed an optimized integer-based similarity computation that reduced computational complexity from $O(n^2)$ to $O(n^{1.585})$, minimizing floating-point operations for energy-efficient AI, IoT, and surveillance applications.

Projects

Netflix ML Analytics Suite | Python, Scikit-learn

2026

- Built and deployed 6 machine learning solutions — recommendation engine, classification models, clustering, time-series forecasting, and an automated analytics pipeline — covering the complete ML lifecycle from feature engineering to hyperparameter tuning (GridSearchCV) across 8+ algorithms, improving model accuracy by up to 10%
- Delivered an automated analytics engine generating data-driven business insights and visual reporting dashboards

Twitter Sentiment Analysis | Python, Scikit-learn, Pandas, NLTK, TF-IDF, Logistic Regression

2024

- Developed a sentiment analysis model using Logistic Regression and TF-IDF vectorization to classify tweets as positive or negative from the Sentiment140 dataset (1.6M tweets).
- Implemented end-to-end NLP preprocessing, including text cleaning, stopwords removal, stemming, feature extraction, model training, and evaluation, achieving 77.8% test accuracy.
- Deployed the trained model using Pickle for efficient sentiment prediction and future integration into NLP applications.

Technical Skills

ML / DL Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, Random Forest, Decision Tree, LSTM, CNN, SVM, Logistic Regression, Transfer Learning, Hyperparameter Tuning

Python & Data Science: Python, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, Google Colab, Streamlit, Gradio, Power BI, Feature Engineering, Data Preprocessing

Computer Vision & NLP: OpenCV, YOLOv3, Object Detection, Non-Maximum Suppression, Feature Extraction, TF-IDF, Tokenization, Sentiment Analysis, Speech Recognition

MLOps & Cloud: Google Cloud Platform, Firebase, Django, REST API, Git, GitHub, Model Deployment

Databases: MySQL, MongoDB, SQLite

Certification

- Winter Certification Program on Prompt Engineering (CSRBOX × IBM SkillsBuild)
- Oracle Certified Data Science Professional (Oracle)
- Oracle Certified Gen-AI Professional (Oracle)